

NOSHIN FOUZIA TASNIM

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[LinkedIn](#) || [Github](#) || [Portfolio](#)

RESEARCH INTEREST

Machine Learning, Image Processing, Biomedical Image Analysis, Human Computer Interaction, Large Language Models

WORK EXPERIENCE

- Lecturer (Contractual), Department of Software Engineering** Sept 2026 - Present
Daffodil International University
- Assigned to Software Engineering Design Capstone Project Course; teaching software development through lectures and coding demonstrations
 - Responsible for mentoring 100 students to plan, build and deploy industry-level, full-stack software projects
- Intern, Product Management, App & Essential Services** Jan 2026 - May 2026
Product & Technology Division, bKash Ltd
- Worked in developing the Knowledge Base with 7500+ data for bKash's virtual assistant, AVA's deployment
 - Contributed to dataset preparation and schema for new AVA conversational flow
 - Helped build AVA's taxonomy for bKash services mapping real user queries from production data
 - Customer Care Center visit to analyse demands of consumers regarding bKash services
- Undergraduate Teaching Assistant/ Student Tutor** Jun 2024 - May 2025
CSE Department, BRAC University
- Assigned to Algorithms course and helped 100+ students during lab classes for assigned section
 - Checked 200+ lab quiz scripts, assignments and mentored students offering 12 hours consultation each week

EDUCATION

- Master of Science in Computer Science and Engineering (Data Science)** Jan 2026 - Present
BRAC University, Dhaka Credits Completed: 9/36
CGPA: 4.00/4.00
- Bachelor of Science in Computer Science and Engineering** Oct 2021 - Nov 2025
BRAC University, Dhaka Credits Completed: 136
CGPA: 3.93/4.00
- Notable Courses: Data Structure & Algorithms, Artificial Intelligence, Data Science, Computer Networks, Computer Architecture, Microprocessors, Software Engineering, Database Systems
- Higher Secondary School Certificate (Science)** 2020
Viqarunnisa Noon School and College, Dhaka GPA: 5.00/5.00
- Secondary School Certificate (Science)** 2018
Viqarunnisa Noon School and College, Dhaka GPA: 5.00/5.00

RESEARCH EXPERIENCE

- Classification of Alzheimer's and Dementia subtypes using R-STDP Driven Spiking Neural Networks**
AI/ML, Neuroimaging, Undergraduate Thesis Repository
- Co-developed an SNN autoencoder model for classification of MRI images of demented and non-demented patients under the supervision of my thesis supervisor, Dr. Md. Golam Rabiul Alam and co supervisor, Zayed Humayun. We built a primary dataset by collecting brain MRI images from the National Institute of Neurosciences for 8 months with IRB approval. My contributions included data collection, preprocessing, building the model and hyperparameter tuning for the best results. The model incorporates Forward-Forward Learning and R-STDP for training without backpropagation and ensemble classifier for testing. It achieved a 87.6% accuracy and F1 score of 87.59% on demented vs non-demented classification.
- Currently preparing the paper for publication in IEEE Journal of Biomedical and Health Informatics
- Usability assessment of e-Tax service platform in Bangladesh across different user groups**
Human Computer Interaction, Ongoing Research
- Conducting research for assessing the usability of e-Return platform for submitting tax return among different user groups in Bangladesh based on age and profession under the supervision of Dr. Farida Chowdhury. We have already conducted heuristic evaluation and are working to evaluate the usability of this platform using different metrics. The data collection process includes user testing and interviewing participants from different user groups, documenting and analyzing user experience and feedback regarding specific task flows of the platform. The user experiences' data will be used to evaluate the usability of the platform.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, Javascript, PHP, SQL, HTML, CSS, RISC-V, x86 assembly language
 - **Frameworks & Libraries :** Laravel, Next.js, PyTorch, Keras, Matplotlib, TensorFlow, Pandas, Scikit-learn
 - **Development Tools:** LaTeX, Git, Github, Postman, Jira, Flex, Bison
 - **Database:** mySQL, PostgreSQL
 - **Hardware design and Networking Tools:** LTSpice, Quartus, Verilog HDL, Cisco Packet Tracer, Wireshark
 - **Microcontrollers:** STM32, Arduino, RaspberryPi
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ACADEMIC PROJECTS

Club Management System

A dynamic website created for club management with deployment

[Repository](#)

- There were different panels for admin, user and clubs with club registration, roles, events, email and notification system
- Next.js framework used, PostgreSQL used for database and Vercel for deployment

A Multi Dimensional Benchmarking study of MSA tools on H1N1 Hemagglutinin Sequences

A benchmarking study of different multisequence alignment tools

[Repository](#)

- The dataset was sampled (n=50-1000) and 4 tools were used: MUSCLE, MAFFT, Clustal Omega & T-Coffee
- The tools were validated against Gold standard OXBench sequences and evaluated across alignment accuracy, computational scalability, phylogenetic consistency and mutation hotspot stability

Smart Water Cleaner Robot

A robot built using Arduino for collecting trash from water and monitoring water quality

[Repository](#)

- Arduino Uno R3 used as microcontroller, C used for arduino code and L293D motor driver shield used for driving motors
- The robot can navigate using HC-SR04, collect trash and monitor water quality using temperature and turbidity sensor and display it on an I2C LCD

Treasure Hunt - Navigating a Maze

A game designed for collecting treasure navigating a maze by breaking bricks and defeating enemies

[Repository](#)

- The game was implemented using OpenGL libraries in Python using only GLPoints
 - Game logic, control and score system were built and display of score, lives and timer
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EXTRACURRICULARS

Assistant Director

Oct 2023 - Dec 2024

Event Management Department, BRAC University Computer Club (BUCC)

- Successfully organized 25+ events, workshops and webinars; led volunteer teams collaborating with other departments
- Managed permissions and received guests and event sponsors; experienced in food allotment and refreshment
- Created a guideline book on venue booking rules in accordance with university policies
- Planned and supervised activities of BUCC Study Corner

Organizer, Intrahacktive 1.0

Oct 2024 - Nov 2024

Event Management Department, BRAC University Computer Club (BUCC)

- Helped with planning and execution of the entire event and organized hackathon, database wizard, debugger
- Led the logistics team and worked with volunteers, guest reception

Executive

Jan 2024 - Dec 2024

BASIS Students Forum BRACU Chapter

- Coordinated between BASIS and BUCC to organize different events and seminars
 - Contributed in meetings and planning for NASA Space Apps Challenge 2024
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HONORS AND AWARDS

• Vice Chancellor's List

For obtaining a GPA of 3.9+ for 9 semesters

• Dean's List

For obtaining a GPA of 3.7+

• Scholarship based on previous results

Received 50% waiver on tuition fees at BRAC University for maintaining a 3.8+ CGPA

• Bronze Award at Duke of Edinburgh International Awards

It was a 6-month long program combining skill development, physical activity, voluntary service and adventurous journey

REFERENCE

Dr. Md. Golam Rabiul Alam

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BRAC University
rabiul.alam@bracu.ac.bd

Dr. Farida Chowdhury

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